2.5 What makes a blockchain contract smart?

Here are some cool features of a smart contract that make it smart. A smart contract is equivalent to any participant in a blockchain network because it has

- A name
- An address
- A cryptocurrency balance (ether, in this case)
- Built-in features to send and receive cryptocurrency (ether, in this case)
- Data and functions
- Built-in features to receive messages and invoke functions
- The ability to reason out the execution of a function

These aspects distinguish and set apart a smart contract from a regular piece of code. A smart contract is indeed smart and different. Let’s examine these features of a smart contract before we begin another application that uses some of these special features.

In a traditional computing system, a participant is identified by a username and an associated password. These elements are used for authentication. This <username, password> combination will not work in a decentralized system, because peers are beyond the usual boundaries of trust (as in the relationships university-student or country-citizen).